



Chen Chen Er

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Email address: tidustide245@gmail.com | **Website:** <https://erchenchen.faculty.bio/> | **LinkedIn:**

<https://www.linkedin.com/in/er-chen-chen-15a38816a/> | **Google Scholar:**

<https://scholar.google.com/citations?user=zU6Dk6QAAAAJ&hl=en> | **Orcid ID:**

<https://orcid.org/0000-0002-0601-9250> | **GitHub:** <https://github.com/pandabear245> |

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Address: August-Bebel-Straße 15B, 01219, Dresden, Germany (Work)

ABOUT ME

Computational Chemist specializing in materials science. Dr. Chen Chen Er's main research interest is on autonomous materials design for energy storage and conversion applications. He is currently based in the Chair of Theoretical Chemistry (Prof. Thomas Heine) at TU Dresden, Germany, as a postdoctoral research fellow under the direct supervision of Dr. Rico Friedrich.

He completed his doctoral studies in Chemical Engineering at Monash University in July 2023 and worked as a postdoctoral research fellow in Singapore from 2023 to 2024. His current research is on modelling and understanding confined high-entropy disordered materials.

WORK EXPERIENCE

RESEARCH FELLOW – TECHNISCHE UNIVERSITÄT DRESDEN – 01/11/2024 – Current – DRESDEN, GERMANY

Business or Sector: Education | **Department:** Professur für Theoretische Chemie | **Email:** chen_chen.er@tu-dresden.de | **Website:** <http://theory.chm.tu-dresden.de/members.shtml?name=ccer>

- Methods development for truly large-scale virtual screening of low-dimensional high-entropy ceramics for extreme materials engineering, in collaboration with Prof. Stefano Curtarolo from Duke University.
- Responsible for developing and implementing software for modeling and predicting the properties of low-dimensional complex disordered systems.

GUEST SCIENTIST – HELMHOLTZ-ZENTRUM DRESDEN-ROSSENDORF – 01/01/2025 – Current – DRESDEN, GERMANY

RESEARCH FELLOW – SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN – 31/08/2023 – 31/05/2024 – SINGAPORE, SINGAPORE

Business or Sector: Professional, scientific and technical activities | **Department:** Science, Mathematics, and Technology Clusters | **Email:** chenchen_er@sutd.edu.sg | **Website:** <https://www.sutd.edu.sg/>

- Developed of large-scale virtual screening workflow for sub-5 nm field-effect transistors design, in collaboration with Prof. Boris Kozinsky from Harvard University.
- Responsible for training machine-learned interatomic potential using neural network approach with NequIP and MACE.

VISITING RESEARCH FELLOW – SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN – 22/04/2023 – 30/08/2023 – SINGAPORE, SINGAPORE

Business or Sector: Professional, scientific and technical activities | **Department:** Science, Mathematics, and Technology Clusters | **Email:** chenchen_er@sutd.edu.sg | **Website:** <https://www.sutd.edu.sg/>

- Developed of large-scale virtual screening workflow for sub-5 nm field-effect transistors design, in collaboration with Prof. Boris Kozinsky from Harvard University.
- Responsible for training machine-learned interatomic potential using neural network approach with NequIP and MACE.

Business or Sector: Education | **Department:** Department of Chemical Engineering | **Email:** Chen.Er@monash.edu | **Website:** <https://www.monash.edu/>

- Developed autonomous workflow for large-scale computational modelling of heterogenous catalysis over green catalysts.
- Responsible for developing and training machine learning models for predicting catalytic pathway with novel green catalysts.

● **EDUCATION AND TRAINING**

02/03/2020 – 26/07/2023 Subang Jaya, Malaysia
PHD. Monash University

- Providing critical insights through fundamental guidance and theoretical evaluations of rational semiconductor design for CO2 reduction applications through photocatalysis using density functional theory (DFT).
- Co-supervised undergraduate research projects in photocatalytic applications.
- Contributed to the setup of Vienna ab initio software simulation package (VASP) and related post-processing software in Monash High Performance Computing (HPC) infrastructure.

Website <https://www.monash.edu/> | **Level in EQF** EQF level 8 | **National classification** Level 10 | **Type of credits** N/A. | **Number of credits** N/A. |

Thesis Mechanistic Insights into the Fundamentals of Photocatalytic Reduction of Carbon Dioxide from First Principles Density Functional Theory Studies
Link https://bridges.monash.edu/articles/thesis/Mechanistic_Insights_into_the_Fundamentals_of_Photocatalytic_Reduction_of_Carbon_Dioxide_from_First_Principles_Density_Functional_Theory_Studies/23690385

02/02/2016 – 19/12/2019 Subang Jaya, Malaysia
BACHELOR IN CHEMICAL ENGINEERING (HONS.) Monash University

Website <https://www.monash.edu/> | **Field of study** Chemical engineering and processes | **Final grade** H2A | **Level in EQF** EQF level 6 | **National classification** Level 9 | **Type of credits** ECTS | **Number of credits** 240

● **LANGUAGE SKILLS**

Mother tongue(s): **CHINESE**
Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C1	C1	C1	C1
MALAY	B2	B2	B2	B2	B2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

● **SKILLS**

Presentation

Microsoft Office

Software

Materials Simulation: LAMMPS, NAMD, VASP | QUANTUM ESPRESSO | Matlab/Simulink

Technical Skills

Molecular modelling with Density-functional theory (DFT) | Reaction Mechanism Study Through Transition State Calculation

Knowledge

Heterogeneous Catalysis | HPC: SLURM, PBS | Operating Systems: UNIX, Linux, Solaris, VxWorks, Windows | Git | Machine Learning | C++ | SQL | Python library used for Scientific computing and technical computing(SciPy) | Python, Scikit-Learn, Numpy, Matplotlib | PyTorch, Keras

Hobby

Enjoy mountain hiking | Playing the piano (ABRSM Grade 5 Certification) | Sports running walking swimming climbing | Good in badminton

PUBLICATIONS

2026

[Dynamic and Resilient Cobalt Copper Nitride and Iron Nitride Heterostructure Aided Energy Economic Urea Electrolysis in Anion Exchange Membrane Electrolyzer](#)

Nibedita Sinha, Chandni Das, **Chen-Chen Er**, Rajashri R Urkude, Santanu Pal, Biplab Ghosh, Siang-Piao Chai, Poulomi Roy (2026). Dynamic and Resilient Cobalt Copper Nitride and Iron Nitride Heterostructure Aided Energy Economic Urea Electrolysis in Anion Exchange Membrane Electrolyzer. Small, e00012. (ISI top 10, JIF: 12.1)

2025

[Interpretable Machine Learning Framework: Predicting Catalytic Activities of Complex Photocatalysts towards CO₂ Conversion](#)

Chen-Chen Er, Lutfi K Putri, Yee Sin Ang, Hui Ying Yang, Siang-Piao Chai (2025). Interpretable machine learning framework: Predicting catalytic activities of complex photocatalysts towards CO₂ conversion. Materials Today Physics, 57, 101800. (ISI top 10, JIF: 9.7)

2025

[Unveiling fundamental first-principles insights into single-atom transition metal photocatalysts for carbon dioxide reduction](#)

Chen-Chen Er, Lutfi K Putri, Yee Sin Ang, Siang-Piao Chai (2025). Unveiling fundamental first-principles insights into single-atom transition metal photocatalysts for carbon dioxide reduction. Fuel, 382, 133746. (JIF: 6.7)

2024

[The Big Problem in Surface Catalysis Modelling](#)

Chen-Chen Er, Yee Wen Teh, Xin Ying Kong, Soungmin Bae, Yu Kumagai, Siang-Piao Chai, Yee Sin Ang (2024). The Big Problem in Surface Catalysis Modelling. ChemRxiv doi:10.26434/chemrxiv-2024-9mbrm.

2024

[Revised Fowler–Dubridge model for multiphoton over-barrier electron emission from two-dimensional materials](#)

Yi Luo, **Chen-Chen Er**, Yee Sin Ang, LK Ang. Revised Fowler–Dubridge model for multiphoton over-barrier electron emission from two-dimensional materials. Applied Physics Letters, 124, 10 (Q2, JIF: 4.0)

2024

[Computational Design of 2D Phosphorus Nanostructures for Renewable Energy Applications: A Review](#)

Chen-Chen Er, Cheng-May Fung, Wei-Kean Chong, Yong Jieh Lee, Lling-Lling Tan, Yee Sin Ang, Nikhil V Medhekar, Siang-Piao Chai (2024). Computational Design of 2D Phosphorus Nanostructures for Renewable Energy Applications: A Review. Advanced Electronic Materials, 10(7), 2300869. (JIF: 5.3)

2024

[MXene quantum dot-sensitized heterostructures for broad solar spectrum CO₂ reduction](#)

Cheng-May Fung, Boon-Junn Ng, Yi-Hao Chew, **Chen-Chen Er**, Jingxiang Low, Xuecheng Guo, Xin Ying Kong, Lling-Lling Tan, Hiroshi Onishi, Abdul Rahman Mohamed, Siang-Piao Chai (2024). MXene quantum dot-sensitized heterostructures for broad solar spectrum CO₂ reduction. Cell Reports Physical Science, 5(11), 102296. (JIF: 7.9)

2023

Effective Low-Powered Photocatalytic Disinfection via Synchronous Introduction of Oxygen Dopants and Carbon Defects in Carbon Nitride

Nur Atika Nikma Dahlan, Lutfi Kurnianditia Putri, **Chen-Chen Er**, Boon-Junn Ng, Chien Wei Ooi, Lling-Lling Tan, Siang-Piao Chai (2023). Effective Low-Powered Photocatalytic Disinfection via Synchronous Introduction of Oxygen Dopants and Carbon Defects in Carbon Nitride. ACS Applied Materials & Interfaces, 15, 46, 53371-53381. (Q1, JIF: 9.5)

2023

Engineering Nanoscale Homo-Heterojunction for Robust Z-Scheme CO₂ Conversion through Synchronous Amalgamation of Oxygen- Defective Ultrathin BiVO₄ and Red/Black Phosphorus

Cheng-May Fung, Boon-Junn Ng, **Chen-Chen Er**, Wei-Kean Chong, Jingxiang Low, Xuecheng Guo, Xin Ying Kong, Hing Wah Lee, Lling-Lling Tan, Abdul Rahman Mohamed, Siang-Piao Chai (2023). Engineering Nanoscale Homo-Heterojunction for Robust Z-Scheme CO₂ Conversion through Synchronous Amalgamation of Oxygen-Defective Ultrathin BiVO₄ and Red/Black Phosphorus. Small Structures, 4, 11, 2300083. (ISI top 10, JIF: 15.9)

2023

Toward Sustainable Ultrawide Bandgap van der Waals Materials: An ab initio Screening Effort

Chuin Wei Tan, Linqiang Xu, **Chen-Chen Er**, Siang-Piao Chai, Boris Kozinsky, Hui Ying Yang, Shengyuan A. Yang, Jing Lu, Yee Sin Ang (2023). Toward Sustainable Ultrawide Bandgap van der Waals Materials: An ab initio Screening Effort. Advanced Functional Materials. (ISI top 10, JIF: 19.0)

2023

Two-dimensional interface engineering of g-C₃N₄/g-C₃N₄ nanohybrid: Synergy between isotype and pn heterojunctions for highly efficient photocatalytic CO₂ reduction

Jie-Yinn Tang, **Chen-Chen Er**, Xin Ying Kong, Boon-Junn Ng, Yi-Hao Chew, Lling-Lling Tan, Abdul Rahman Mohamed, Siang-Piao Chai (2023). Two-dimensional interface engineering of g-C₃N₄/g-C₃N₄ nanohybrid: Synergy between isotype and pn heterojunctions for highly efficient photocatalytic CO₂ reduction. Chemical Engineering Journal, 466, 143287. (ISI top 10, JIF: 15.1)

2023

Sensitization of Bismuth Tungstate with Magnesium Phthalocyanine for Photoinduced Charge Redistribution and Electron Localization: CO₂ Photoreduction from Vis to NIR Region

Yee Wen Teh, Wei Kean Chong, **Chen-Chen Er**, Xin Ying Kong, Boon-Junn Ng, Siek Ting Yong, Siang-Piao Chai (2023). Sensitization of Bismuth Tungstate with Magnesium Phthalocyanine for Photoinduced Charge Redistribution and Electron Localization: CO₂ Photoreduction from Vis to NIR Region. Journal of Materials Chemistry C, 11, 705711. (Q1, JIF: 6.4)

2022

Insights from density functional theory calculations on heteroatom P-doped ZnIn₂S₄ bilayer nanosheets with atomic-level charge steering for photocatalytic water splitting

Wei-Kean Chong, Boon-Junn Ng, **Chen-Chen Er**, Lling-Lling Tan, Siang-Piao Chai (2022). Insights from density functional theory calculations on heteroatom P-doped ZnIn₂S₄ bilayer nanosheets with atomic-level charge steering for photocatalytic water splitting. Scientific Reports, 12, 1927. (Q2, JIF: 4.6)

2022

Unraveling the roles of single transition metal atom anchored on equivalent stoichiometry graphitic carbon nitride (gC₆N₆) for carbon dioxide reduction: a density functional theory study

Chen-Chen Er, Lutfi K. Putri, Nikhil V. Medhekar, Siang-Piao Chai (2022). Unraveling the roles of single transition metal atom anchored on equivalent stoichiometry graphitic carbon nitride (gC₆N₆) for carbon dioxide reduction: a density functional theory study. Journal of Physics D: Applied Physics, 56(2), 024004. (Q2, JIF: 3.4)

2022

Synergistic Hybridization of Twin-Induced Red/Black Phosphorus and Tungsten Oxide as Homo-Hetero Dynamic Dual Junctions for Z-Scheme CO₂ Photoreduction

Cheng-May Fung, Boon-Junn Ng, **Chen-Chen Er**, Xin Ying Kong, Lling-Lling Tan, Abdul Rahman Mohamed, Siang-Piao Chai (2022). Synergistic Hybridization of Twin-Induced Red/Black Phosphorus and Tungsten Oxide as Homo-Hetero Dynamic Dual Junctions for Z-Scheme CO₂ Photoreduction. ACS Applied Energy Materials, 5, 12, 15257–15268. (Q1, JIF: 6.4)

2022

Allotropes selection apropos of photocatalytic CO₂ reduction from first principles studies

Chen-Chen Er, Lutfi K. Putri, Boon-Junn Ng, Nikhil V. Medhekar, Siang-Piao Chai (2022). Allotropes selection apropos of photocatalytic CO₂ reduction from first principles studies. Materials Today Physics, 26, 100751. (ISI top 10, JIF: 11.5)

2022

Charge Modulation at Atomic-Level through Substitutional Sulfur Doping into Atomically Thin Bi₂WO₆ toward Promoting Photocatalytic CO₂ Reduction

Yee Wen Teh, **Chen-Chen Er**, Boon-Junn Ng, Xin Ying Kong, Siek-Ting Yong, Siang-Piao Chai (2022). Charge Modulation at Atomic-Level through Substitutional Sulfur Doping into Atomically Thin Bi₂WO₆ toward Promoting Photocatalytic CO₂ Reduction. ChemSusChem 15(14) e202200471. (Q1, JIF: 8.4)

2022

Uncovering the multifaceted roles of nitrogen defects in graphitic carbon nitride for selective photocatalytic carbon dioxide reduction: a density functional theory study

Jie-Yinn Tang*, **Chen-Chen Er***, Lling-Lling Tan, Yi-Hao Chew, Abdul Rahman Mohamed, and Siang-Piao Chai (2022). Uncovering the multifaceted roles of nitrogen defects in graphitic carbon nitride for selective photocatalytic carbon dioxide reduction: a density functional theory study. (*Both authors contributed equally) Physical Chemistry Chemical Physics, 24 (18), 11124-11130. (Q1, JIF: 3.3)

2021

Red Phosphorus: An Up-and-Coming Photocatalyst on the Horizon for Sustainable Energy Development and Environmental Remediation

Cheng-May Fung, **Chen-Chen Er**, Lling-Lling Tan, Abdul Rahman Mohamed, Siang-Piao Chai (2022). Red Phosphorus: An Up-and-Coming Photocatalyst on the Horizon for Sustainable Energy Development and Environmental Remediation. Chemical Reviews. 122, 3, 3879-3965. (ISI top 10, JIF: 62.1)

2021

Atomistic Insights into the Reformation of CH₄ with CO₂ on Metal-Free gC₃N₄: Unraveling the Reaction Mechanisms Using First-Principles DFT Calculations

Chen-Chen Er, Jie-Yinn Tan, Nikhil V. Medhekar, Siang-Piao Chai (2021). Atomistic Insights into the Reformation of CH₄ with CO₂ on Metal-Free gC₃N₄: Unraveling the Reaction Mechanisms Using First-Principles DFT Calculations. Journal of Physical Chemistry C, 125(42), 23021-23028. (Q2, JIF: 3.7)

2021

Proton-functionalized graphitic carbon nitride for efficient metal-free disinfection of Escherichia Coli under low-power light irradiation

Boon-Junn Ng, Muhammad Khosyi Musyaffa, **Chen-Chen Er**, Kulandai Arockia Rajesh Packiam, WP Cathie Lee, Lling-Lling Tan, Hing Wah Lee, Chien Wei Ooi, Siang-Piao Chai (2021). Proton-functionalized graphitic carbon nitride for efficient metal-free disinfection of Escherichia Coli under low-power light irradiation. Chemistry–A European Journal, 27(9), 3085-3090. (Q2, JIF: 4.3)

2020

Insights on the Impact of Doping Levels in Oxygen-doped gCN and its Effects on Photocatalytic Activity

Lutfi K. Putri, Boon-Junn Ng, **Chen-Chen Er**, Wee-Jun Ong, Wei Sea Chang, Abdul Rahman Mohamed, Siang-Piao Chai (2020). Insights on the Impact of Doping Levels in Oxygen-doped gCN and its Effects on Photocatalytic Activity. Applied Surface Science, 504, 144427. (ISI top 10, JIF: 6.7)

2020

Tuning the Electronic Band Structure of Graphitic Carbon Nitride by Breaking Intramolecular Bonds: A Simple and Effective Approach for Enhanced Photocatalytic Hydrogen Production

Michell KT Chee, **Chen-Chen Er**, Jie-Yinn Tang, Lling-Lling Tan, Wei Sea Chang, Siang-Piao Chai (2020). Tuning the Electronic Band Structure of Graphitic Carbon Nitride by Breaking Intramolecular Bonds: A Simple and Effective Approach for Enhanced Photocatalytic Hydrogen Production. Applied Surface Science, 529, 146600. (ISI top 10, JIF: 6.7)

● NETWORKS AND MEMBERSHIPS

01/05/2020 – CURRENT Malaysia

Boards of Engineers Malaysia (BEM)

Graduate Engineer
Membership No.: G1182180A

01/02/2020 – CURRENT United Kingdoms

Institution of Chemical Engineers (IChemE)

Associate Member
Membership No.: 100216623

01/04/2026 – CURRENT

Collaborative Research Centre (CRC) 1415 "Chemistry of Synthetic Two-Dimensional Materials"

Associate member
Project C08
Functional 2D Heterostructures by Data-Driven Design
Rico Friedrich – Principal Investigator

● CONFERENCES AND SEMINARS

16/03/2025 – 21/03/2025 Regensburg, Germany.

Deutsche Physikalische Gesellschaft (DPG)-Frühjahrstagungen 2025

Poster Contribution: Modelling and Descriptor-based Synthesizability Analysis of High-Entropy Materials.

Link <https://regensburg25.dpg-tagungen.de/programm/assets/r25book.pdf>

2023 – 2023 Singapore

1st- International Conference on Low-Energy Digital Devices and Computing (ICLED-2023)

2023 – 2023 Singapore

Institute of Physics Singapore (IPS) 2023

2022 – 2022 Malaysia

Chulalongkorn University Thailand – Monash University Malaysia Bilateral Research Seminar on Catalysis 2022

● HONOURS AND AWARDS

02/03/2020

Graduate Research Tuition Fee Waiver Scholarship – Monash University

100% tuition fee waiver up to the duration stated below:

- Doctoral degree - Three years, six months full-time equivalent from the date of course commencement until the scholarship end date or upon thesis submission; whichever is earlier.

Co-investigator, SUTD Kickstarter initiative (Award No: SKI 2021_04_01) – Singapore University of Technology and Design

SGD \$100,000

Topic: "Designing Nanostructure with Enhanced Charge Storing Capacity for Round-the-Clock Photocatalytic Energy Conversion"

National Supercomputing Centre (NSCC) 2nd Call for Projects Jan'24 Cycle – NSCC Singapore

(Project ID:15003684) **3,750,000 CPU hours**

Topic: "High-throughput screening of 2D nanomaterials for persistent photocatalysis."

● **TEACHING TRACK**

01/02/2024 – 31/12/2024

Science for a Sustainable World [SUTD]

Adjunct Teaching Fellow

-This course focuses on the importance of sustainable development by creating awareness of climate change and the human impact on the environment. Based on the Sustainable Development Goals (SDGs) from the United Nations, this course will address issues and challenges related to sustainability in four main areas: green materials, clean water, sustainable energy, and biodiversity. Emphasis is placed on the knowledge of science and engineering approaches towards achieving sustainable development. This course aims to enhance students' critical thinking skills in analyzing strategies based on scientific facts and current technologies.

01/12/2023 – 31/12/2023

Compact Physical World for SUTD-Polytechnic Special Arrangement [SUTD]

Ad hoc Teaching Fellow

-Emphasis on providing students with the ability to understand and explain the inner mechanism of the physical world based on the principles of mechanics and thermodynamics. It aims to help students appreciate the beauty of physics and enable them to apply key concepts learnt to evaluate and address physics-based problems to make a positive impact on the world. By using concepts established through simplified mathematical models, reverse engineering case studies and experiential learning through hands-on demonstrations, connections between physics concepts and theoretical models are reinforced with practice.

● **REFERENCES**

Prof. Ir. Dr. Chai Siang Piao

Deputy Head of School (Research)
School of Engineering
Monash University Malaysia
47500 Subang Jaya
Tel: +603 5514 6234
Email: chai.siang.piao@monash.edu

A/Prof. Dr. Ng Boon Junn

Assistant Professor
School of Energy and Chemical Engineering
Xiamen University Malaysia
Jalan Sunsuria, Bandar Sunsuria, 43900 Sepang,
Selangor, Malaysia
Tel: N/A.
Email: boonjunn.ng@xmu.edu.my

Prof. Dr. Nikhil V. Medhekar

Department of Materials Science and Engineering
Faculty of Engineering
Monash University Clayton

Victoria 3800
Tel: +613 990 51421
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